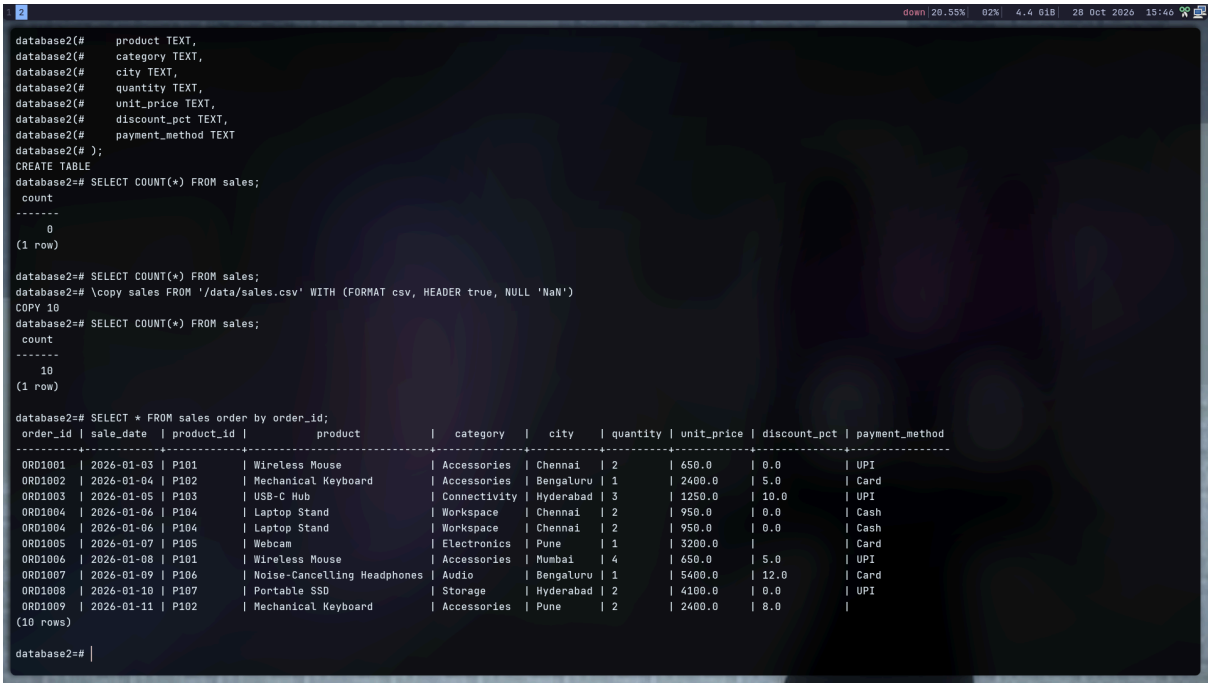
Schema

```
CREATE TABLE sales (
    order_id TEXT,
    sale_date TEXT,
    product_id TEXT,
    product TEXT,
    category TEXT,
    city TEXT,
    quantity TEXT,
    unit_price TEXT,
    discount_pct TEXT,
    payment_method TEXT
);
```

Copy cmd for csv file

```
\copy sales FROM '/data/sales.csv' WITH (FORMAT csv, HEADER true, NULL 'NaN')
```

1. **Retrieve all the records**



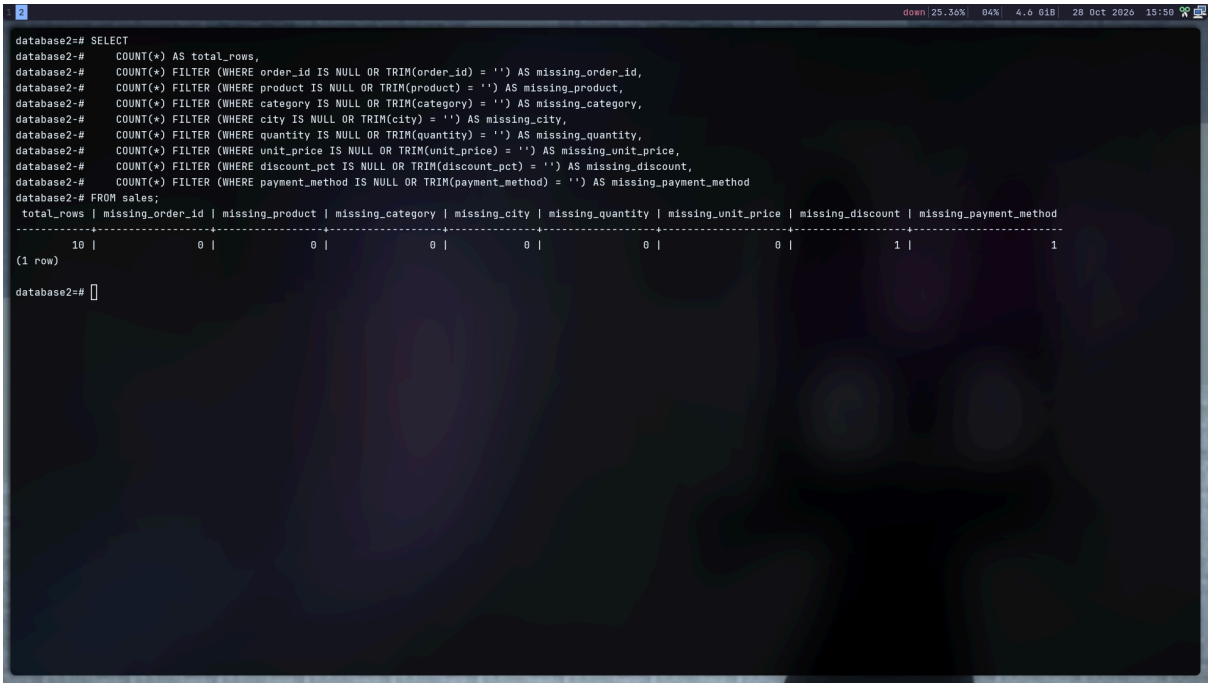
```
database2(#     product TEXT,
database2(#     category TEXT,
database2(#     city TEXT,
database2(#     quantity TEXT,
database2(#     unit_price TEXT,
database2(#     discount_pct TEXT,
database2(#     payment_method TEXT
database2(# );
CREATE TABLE
database2=# SELECT COUNT(*) FROM sales;
 count
-------
     0
(1 row)

database2=# SELECT COUNT(*) FROM sales;
database2=# \copy sales FROM '/data/sales.csv' WITH (FORMAT csv, HEADER true, NULL 'NaN')
COPY 10
database2=# SELECT COUNT(*) FROM sales;
 count
-------
    10
(1 row)

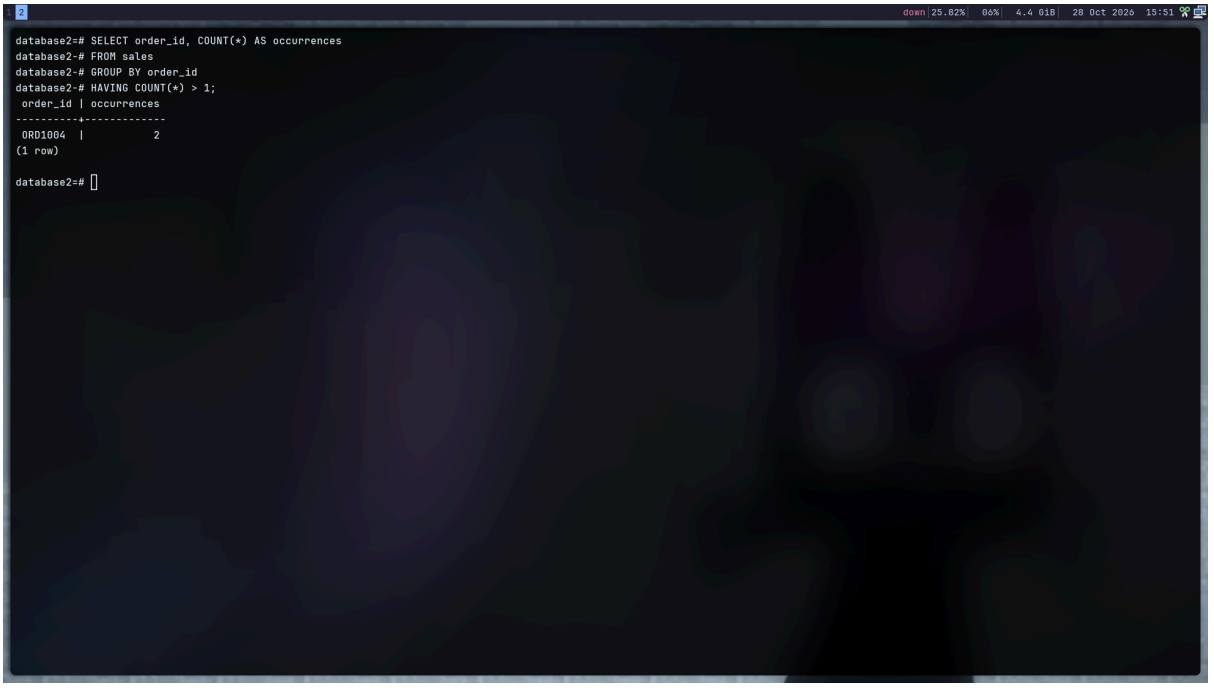
database2=# SELECT * FROM sales order by order_id;
 order_id | sale_date  | product_id |          product           |   category   |   city    | quantity | unit_price | discount_pct | payment_method
----------+------------+------------+----------------------------+--------------+-----------+----------+------------+--------------+----------------
 ORD1001  | 2026-01-03 | P101       | Wireless Mouse             | Accessories  | Chennai   | 2        | 650.0      | 0.0          | UPI
 ORD1002  | 2026-01-04 | P102       | Mechanical Keyboard        | Accessories  | Bengaluru | 1        | 2400.0     | 5.0          | Card
 ORD1003  | 2026-01-05 | P103       | USB-C Hub                  | Connectivity | Hyderabad | 3        | 1250.0     | 10.0         | UPI
 ORD1004  | 2026-01-06 | P104       | Laptop Stand               | Workspace    | Chennai   | 2        | 950.0      | 0.0          | Cash
 ORD1004  | 2026-01-06 | P104       | Laptop Stand               | Workspace    | Chennai   | 2        | 950.0      | 0.0          | Cash
 ORD1005  | 2026-01-07 | P105       | Webcam                     | Electronics  | Pune      | 1        | 3200.0     |              | Card
 ORD1006  | 2026-01-08 | P101       | Wireless Mouse             | Accessories  | Mumbai    | 4        | 650.0      | 5.0          | UPI
 ORD1007  | 2026-01-09 | P106       | Noise-Cancelling Headphones | Audio        | Bengaluru | 1        | 5400.0     | 12.0         | Card
 ORD1008  | 2026-01-10 | P107       | Portable SSD               | Storage      | Hyderabad | 2        | 4100.0     | 0.0          | UPI
 ORD1009  | 2026-01-11 | P102       | Mechanical Keyboard        | Accessories  | Pune      | 2        | 2400.0     | 8.0          |
(10 rows)

database2=#
```

2. **Check missing values**



3. Find Duplicate order ids

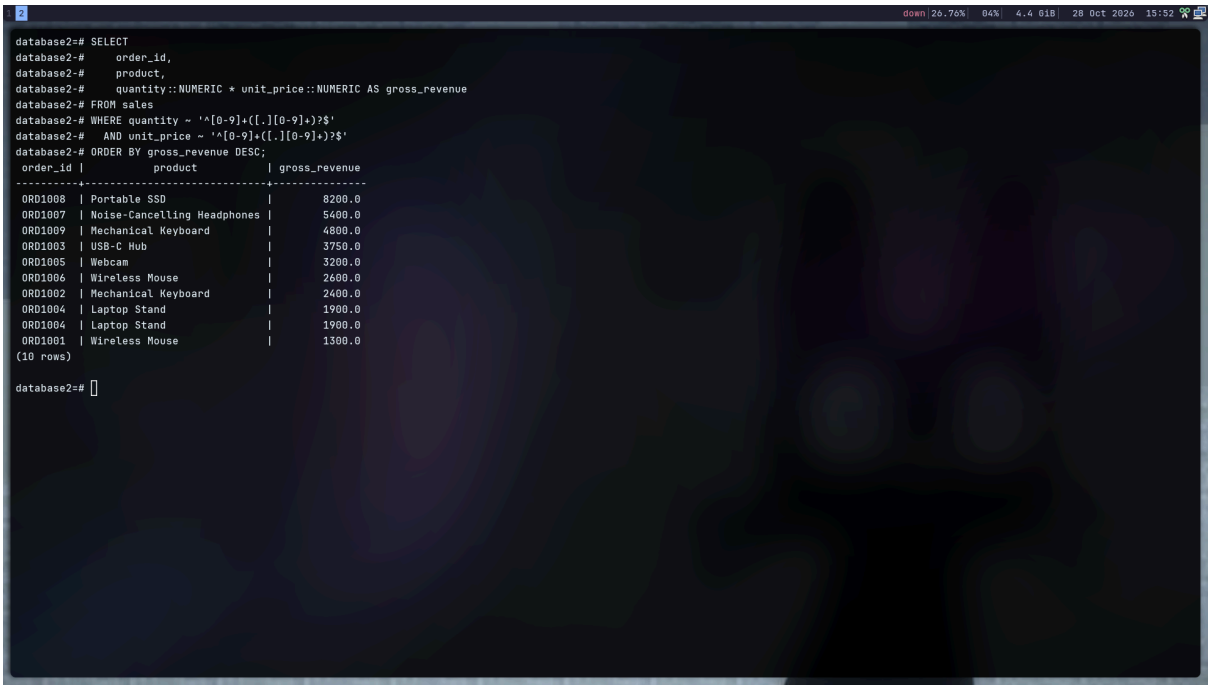


z

4. **Calculate revenue before discount**

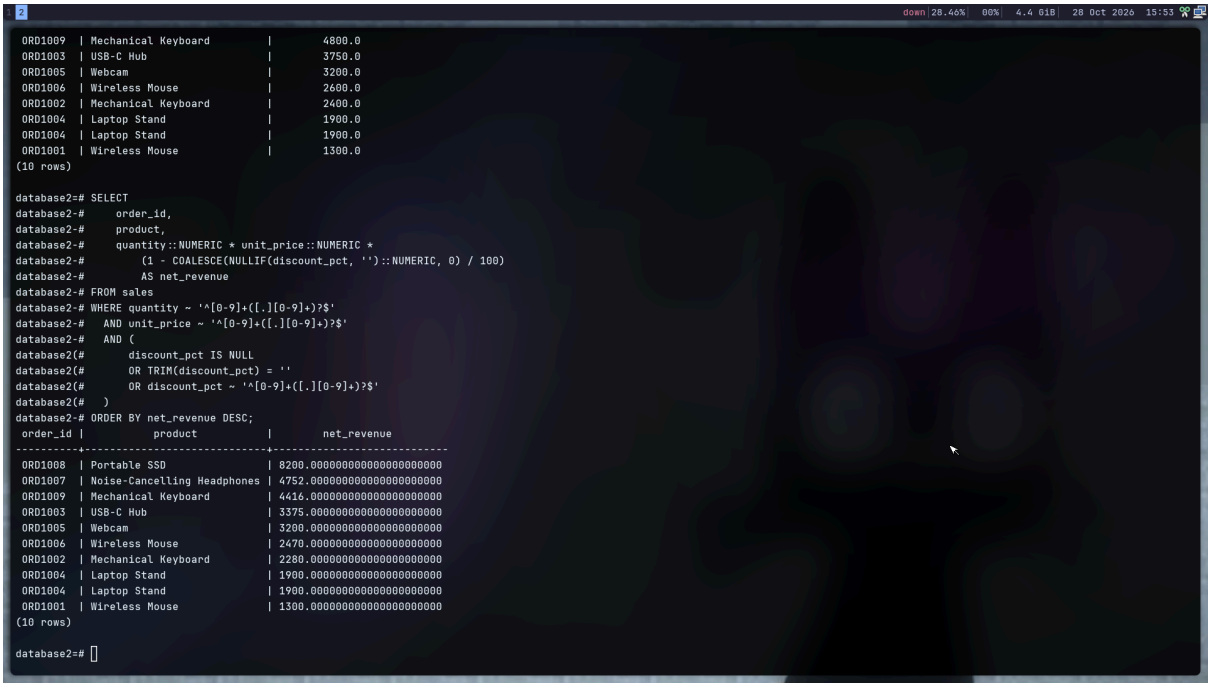
Query:

```
SELECT
    order_id,
    product,
    quantity::NUMERIC * unit_price::NUMERIC AS gross_revenue
FROM sales
WHERE quantity ~ '^[0-9]+([.][0-9]+)?$'
  AND unit_price ~ '^[0-9]+([.][0-9]+)?$'
ORDER BY gross_revenue DESC;
```



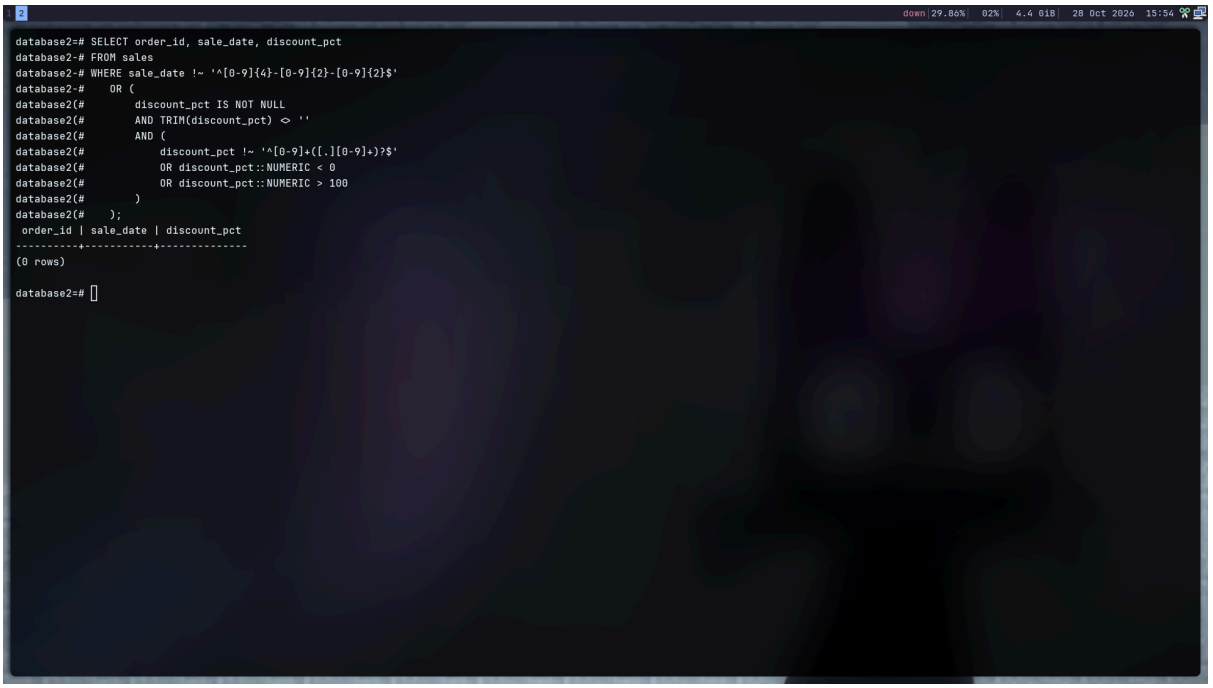
5. **Calculate revenue after discount**

```
Query:SELECT
    order_id,
    product,
    quantity::NUMERIC * unit_price::NUMERIC *
        (1 - COALESCE(NULLIF(discount_pct, '')::NUMERIC, 0) / 100)
        AS net_revenue
FROM sales
WHERE quantity ~ '^[0-9]+([.][0-9]+)?$'
  AND unit_price ~ '^[0-9]+([.][0-9]+)?$'
  AND (
      discount_pct IS NULL
      OR TRIM(discount_pct) = ''
      OR discount_pct ~ '^[0-9]+([.][0-9]+)?$'
  )
ORDER BY net_revenue DESC;
```



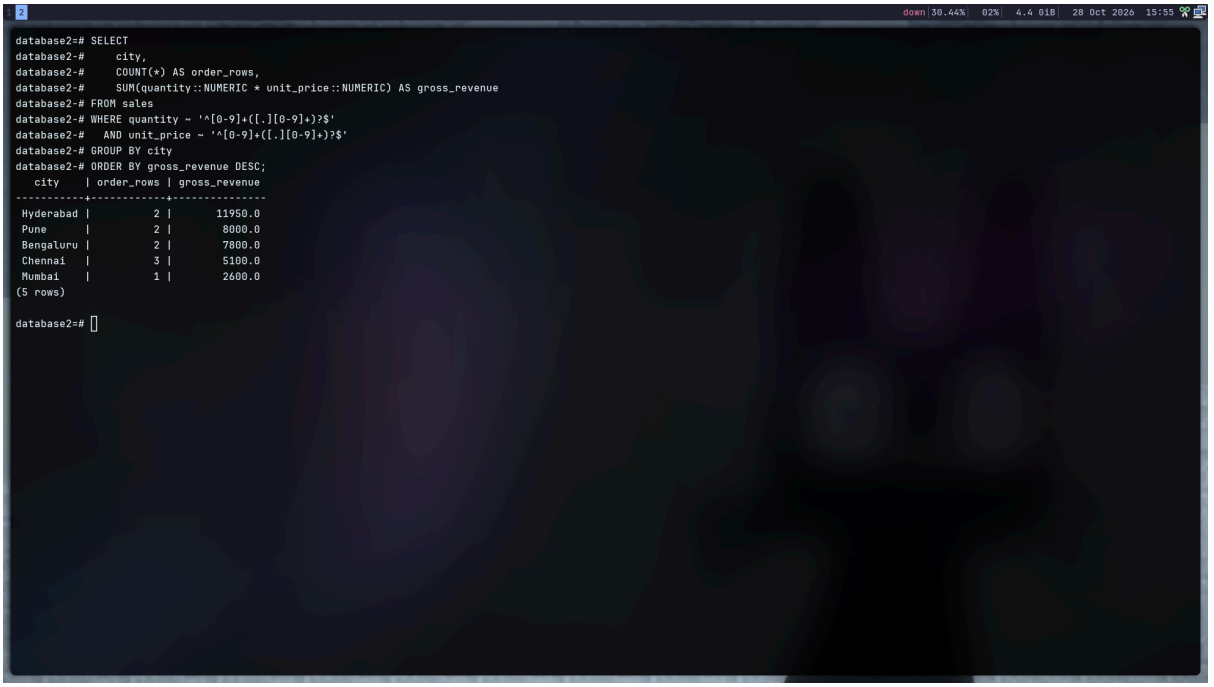
6. **Check date validity and unusual discount**

```
SELECT order_id, sale_date, discount_pct
FROM sales
WHERE sale_date !~ '^[0-9]{4}-[0-9]{2}-[0-9]{2}$'
  OR (
    discount_pct IS NOT NULL
    AND TRIM(discount_pct) <> ''
    AND (
      discount_pct !~ '^[0-9]+([.][0-9]+)?$'
      OR discount_pct::NUMERIC < 0
      OR discount_pct::NUMERIC > 100
    )
  );
```



7. **Group gross revenue by city**

```
SELECT
    city,
    COUNT(*) AS order_rows,
    SUM(quantity::NUMERIC * unit_price::NUMERIC) AS gross_revenue
FROM sales
WHERE quantity ~ '^[0-9]+([.][0-9]+)?$'
  AND unit_price ~ '^[0-9]+([.][0-9]+)?$'
GROUP BY city
ORDER BY gross_revenue DESC;
```


1 2
down 30.44% 02% 4.4 GiB 28 Oct 2026 15:55
database2=# SELECT
database2-#   city,
database2-#   COUNT(*) AS order_rows,
database2-#   SUM(quantity::NUMERIC * unit_price::NUMERIC) AS gross_revenue
database2-# FROM sales
database2-# WHERE quantity ~ '^[0-9]+([.][0-9]+)?$'
database2-#   AND unit_price ~ '^[0-9]+([.][0-9]+)?$'
database2-# GROUP BY city
database2-# ORDER BY gross_revenue DESC;
city | order_rows | gross_revenue
Hyderabad | 2 | 11950.0
Pune | 2 | 8000.0
Bengaluru | 2 | 7800.0
Chennai | 3 | 5100.0
Mumbai | 1 | 2600.0
(5 rows)
database2=#